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A Pangenome Sequence Search and Coordinate Translation Service for the UCSC Genome Browser

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poster sources
parsaeskandar/hprc_2026_poste

Paste a **sequence** or a **position** — get an answer from **every HPRC haplotype**, live, inside the browser.

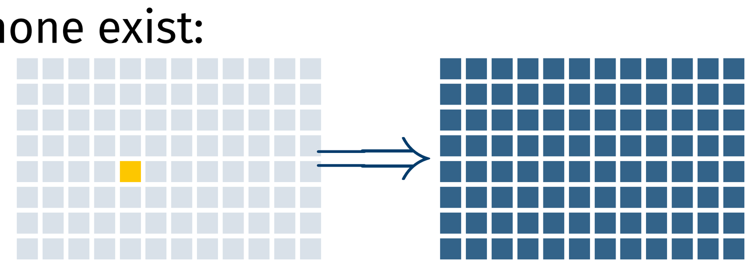
464 haplotypes scored per query

~20 ms gene-scale liftover

466 × 466 assembly pairs, on demand

01 The graph nobody could ask

The HPRC v2.0 graph holds **233 samples / 464 haplotypes**, yet there has been no interactive way to query it. Liftover needs a chain file per assembly pair — and almost none exist:



A researcher with a **sequence** can't ask which haplotypes carry it; one with a **position** can't move it between assemblies. We built the service that answers both, inside the UCSC Genome Browser.

02 What users get

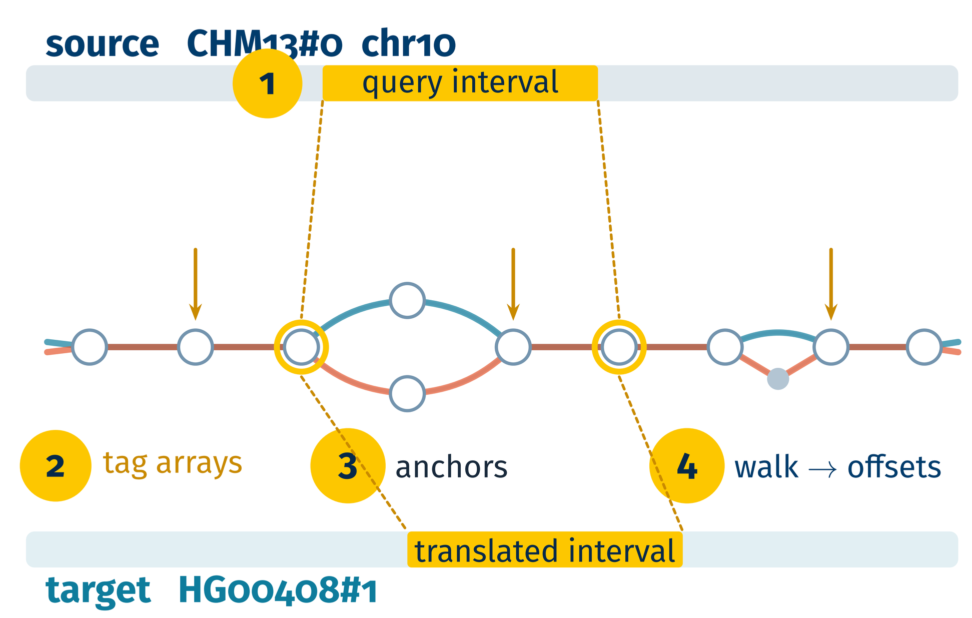
Pangenome sequence search
A BLAT-like page: a pasted sequence is mapped to the whole graph with `vg giraffe`; results rank all 464 haplotypes by identity and link to its position on any of them, drawn as a track with base-level mismatch coloring.

Any-to-any coordinate translation
Liftover between any two of 466 assemblies at interactive speed, with a scored "which assemblies contain this region" destination picker.

Annotation lifting
Users land on an HPRC assembly with CHM13's genes, RefSeq, CenSat and T2T Encode at translated positions, plus an **Alignment Differences** track — via a QuickLift chain generated on the fly.

03 One index, every haplotype

Everything rides on the **tag-array index** (Sirén et al.) built once over the HPRC graph: it annotates **every node** with the haplotypes that cross it, and where. Translation is a walk through the graph — not a lookup in 213,000 pairwise files:



1 Locate the query interval on its graph path. **2** The tag array answers, per node, *every* haplotype standing on it — all 464 in one pass. **3** Nodes visited exactly once by source and target are guaranteed orthologous anchors. **4** Walking between them emits per-base offsets, folded into chain blocks.

The same pass **scores a mapped read against all 464 haplotypes**:

Shown together, the two scores separate cases one number conflates:

- 100 / 99 present, near-exact
- 100 / 60 present, diverged
- 50 / 50 half missing

04 In the browser

genome.ucsc.edu

add figures/mapping_page.png
Pangenome Mapping page: identity-ranked assemblies, position-on-haplotype links

One search scores every haplotype; picking one **resurjects the cached alignment** — no remapping — and the browser opens there with the sequence as a native track:

genome.ucsc.edu

add figures/pangenome_seq_track.png
"Pangenome Seq" track, BLAT-style base-level mismatch coloring

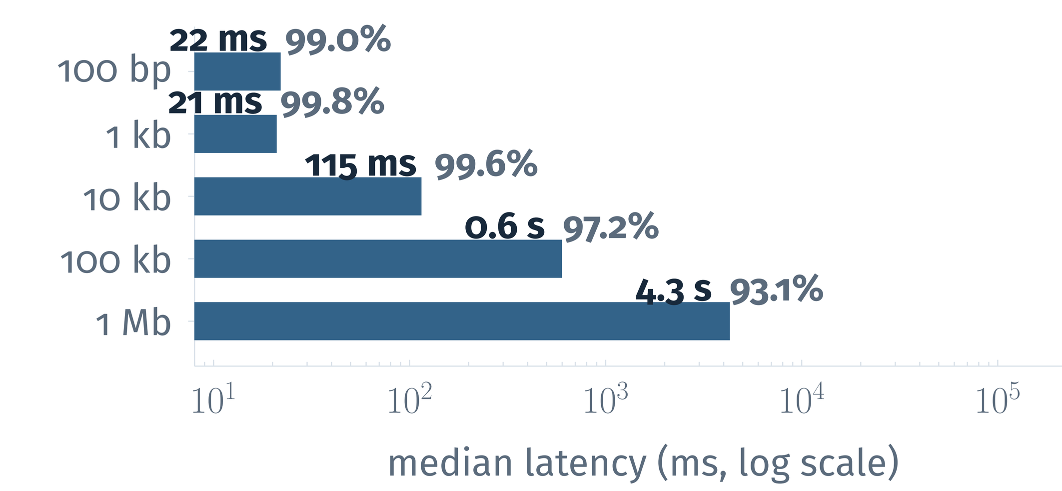
Converting a position lands on the target with **CHM13's annotations lifted** and an **Alignment Differences** track — across 22 kb of ABO it shows exactly the 2 real C→T mismatches between hs1 and HG00597#1:

genome.ucsc.edu

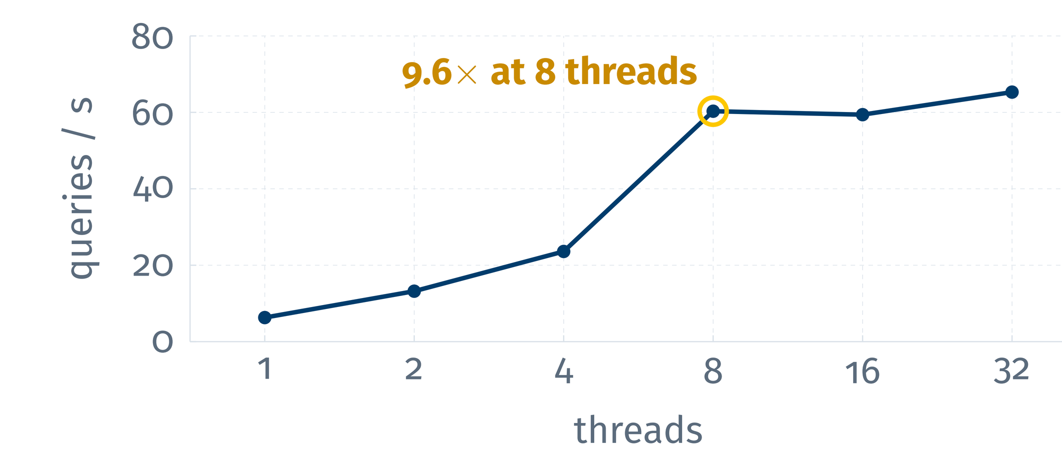
add figures/lifted_annotations.png
HPRC assembly with lifted CHM13 tracks + Alignment Differences

05 Fast enough to be interactive

Translation latency (completeness in gray):



Concurrent queries (3.4 kb):



Fast enough that search, translation and annotation lifting all run at page-load speed, for any pair of the 466 assemblies.

06 What's next

Public release
Bringing pangenome sequence search and any-to-any coordinate translation to everyone, in the public UCSC Genome Browser.

07 Acknowledgements

We thank our colleagues in the UC Santa Cruz Computational Genomics Lab — especially **Adam Novak**, **Glenn Hickey**, and **Zia Truong** — and the UCSC Genome Browser team. Built on `vg` and the HPRC v2.0 graph.